Comparing schemes and Manifolds

This is just a quick extract from [1] that I wanted to post that compares schemes and smooth manifolds. It is not comprehensive, it is just the first things that came to mind. For a full discussion, see the cited notes.

1. The scheme given by $V(x)\cup V(yz) \subseteq \mathbb{A}^3_k$ has two irreducible components of different dimensions. A manifold (differential and even topological) will always be of a single dimension.

2. A manifold cannot self-intersect[1], while schemes (even affine schemes, even varieties) can: consider $V(y^2 - x^3 - x^2) \subseteq \mathbb{A}^2_k$.

3. The patches of manifold are very regular, being diffeomorphic to $\mathbb{R}^n$, however patches of a scheme may be "pathological", for example if the ring is not Noetherian then there may be uncountably many connected components. Some of the conditions and properties we shall introduce later will add some regularity to the components.

4. In Manifold Theory, the notion of irreducible component is trivial as only points are irreducible. From an analytic perspective, this can be thought of as the lack of separability of points by functions for schemes; part of this consequence is the existence of "large" points for schemes

5. If a scheme X is irreducible, all its open sets are dense. Many of the schemes we shall work with will be the union of irreducible subschemes, where for each irreducible subscheme all affine open subsets are dense in the irreducible subscheme; having dense open subsets is far from being the case for most manifolds.

6. More generally, the behaviour of schemes matches more closely with the behaviour of complex geometry (we shall draw such analogies often, for example *Hartog's Extension Lemma* ([3, chapter 8]) shall have parallels with normal schemes, the existence of certain holomorphic/meromorphic functions shall tie into divisor theory, compact complex Riemann surfaces shall correspond to projective curves, and so forth).

7. Schemes will be able to have "double points", recall the line with two origin is a popular counter for a manifold as it is not Hausdorff at the origin.

8. A scheme remembers multiplicity. For example, we can define a lot of functions on a single point, however scheme functions (i.e. regular functions) can hold multiplicity information (ex. $\operatorname{Spec} k[x]/(x)^2$). A smooth manifold $(M, \mathcal{A})$ with maximal atlas does not hold this information.

9. There can be multiple structure sheaves on the same set defining different schemes (think of the different fields that can form the sheaf over a singleton). Manifolds of dimension ≤ 3 have unique differential structures up to diffeomorphism.

10. Though there wasn't an atlas defined for schemes (as the notion of maximal atlas is not as relevant), schemes can be constructed via gluing just like an atlas on a manifold.

[1] Naturally, a submersion of a manifold may have intersections, however (naturally) the immersion is not a manifold

11. the natural additional structure on a smooth manifold is a *tangent bundle* with transition mappings of the form $D(\mathbf{y} \circ \mathbf{x}^{-1})$, namely they are not just smooth transitions between open sets and fibers, but need to respect a $\mathrm{GL}_n(\mathbb{R})$-action given by the derivation D. There is an equivalent *tangent sheaf* for schemes, but more care must be done as schemes are not "locally linear" everywhere in the way that smooth manifolds must be. Much of the intuition of Poincaré's duality (see [2, section 4.3]) will influence the generalization.

12. Scheme functions are more "rigid"; while on a manifold we may have partitions of unity due to the flexibility of euclidean topology, no exact equivalent can be defined for schemes. This rigidity will also add structure and make it easier to defined some concepts (ex. closed subschemes shall be a rather "natural" concept as compared to the work that goes into ensuring closed submanifolds are locally isomorphic to $\mathbb{R}^k$).

    As smooth functions on each $U \subseteq M$ from a sheaf, sheaves that admit a "partition of unity" will be called *fine sheaves*. A weaker notion of *soft sheaf* will be the closest equivalent to having the partition of unity property.

13. Schemes unify "arithmetic" and "geometry" in a way manifold theory is poorly equipped for. As an example, there is no equivalent of $\operatorname{Spec}\mathbb{Z}$, $\operatorname{Spec}\mathcal{O}_K$ or $\operatorname{Spec}\mathbb{Z}[t]$ where $\mathcal{O}_K$ is a ring of integers for smooth manifolds. In this way, scheme theory also allows for a new type of object that is not tractable in differential geometry.

14. very often in practice (especially in the setting of k-schemes and study curves, surfaces, moduli spaces, etc.) we work with integral separated schemes that most of the time are proper, and we have many methods to "normalize" singularities (ex. Intersections, cusps, etc) to make these scheme regular[2]. In fact, a result known as *GAGA* shows when we can interpret proper schemes with appropriate coherent sheaves as complex manifolds. Naturally, arithmetic schemes will not be part of this manifold/scheme correspondence.

References

[1] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Geometry*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algGeo.pdf.

[2] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Topology*. 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebraic_topology.pdf.

[3] Nathanael Chwojko-Srawley. *Everything You Need To Know About Complex Analysis*. 1st ed. NA. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_complex_analysis.pdf.

[2]Though as we'll see, there is no universal way of doing this with the exception of some low-dimension that have been completely proved